Code Generation

Introduction

- Code generation uses techniques studied so far such as fine-tuning, retrieval-augmented generation, and large-scale language models.
- We will look at 4 systems:
  - StarCoder
  - REDCODER
  - Yu et al.'s evaluation of ChatGPT
  - CodeBLEU

StarCoder

- Open-access Code LLM by BigCode.

- 15.5B parameters
- 8K context length.

- Trained on 1T tokens from GitHub's permissive code.

- Supports 80+ programming languages.

StarCoder Architecture

- Pretrained version before fine-tuning.

- Fine-tuned on 35B Python tokens.
- Outperforms open Code LLMs.
- Matches or beats OpenAI's code-cushman-001 (Microsoft Copilot).
- Uses Multi-Query Attention (MQA) for fast inference.
- Includes Fill-in-the-Middle (FIM) capabilities.

StarCoder Safety and Privacy

- Safe release with privacy protections.
- Implements Personal Identifying Information (PII) redaction pipeline.

- Includes an attribution tool for transparency.
- Released under OpenRAIL-M license.

StarCoder Datasets

- Covers languages like Python, Java, C, Rust.
- Built on **The Stack** dataset.
  - Allows community opt-out from training data.
- Reduces carbon footprint vs. closed models.

StarCoder Evaluation

- Extensive evaluations on coding benchmarks.
  - Superior on HumanEval, MBPP, DS-1000.
  - Better than LLaMA and other previous systems
- Fine-tuning improves Python performance significantly.

REDCODER

- Retrieval-augmented code generation and summarization.

- Mimics developer behavior by retrieving relevant code/snippets.

- Retrieves from a code database to assist generation.

SCODE-R

- A dense retrieval model (SCODE-R).
- Improves retrieval over BM25.
- Retrieves both code and summaries.
- Retrieval database includes GitHub, Stack Overflow.

SCODE-G

- Generates code/summaries from retrieved content.
- Based on PLBART transformer model.

RECODER Evaluation

- Evaluated on Java and Python benchmarks.

  - Outperforms prior models on BLEU, CodeBLEU, and Exact Match.
- Retrieval improves code generation accuracy.
- Retrieval improves summarization accuracy.
- Fine-tuned SCODE-R surpasses sparse retrieval methods.
- Human evaluation confirms competitive code quality.

ChatGPT Evaluation

- Yu et al. evaluate ChatGPT for code generation.
- Focus on code generation, completion, and repair.
- ChatGPT acts as both developer and tester.
- Self-verifies generated code using test reports.

Reliability of ChatGPT

- Often misjudges incorrect code as correct.
- Fails to detect vulnerabilities in completed code.
- Incorrectly labels failed repairs as successful.

- Self-contradicts by later predicting issues in its own code.

ChatGPT Evaluation

- Three self-verification methods used:
  - direct question,
  - guiding question - improves error detection,
  - test report - better identifies vulnerabilities..
- Explanations in test reports often inaccurate.
- Hallucinations lead to inconsistent self-verification.

ChatGPT As Developer

- Findings highlight ChatGPT's limitations in self-verification.
- Developers must manually verify generated code.
- Self-verification performance varies by programming language.
- Better prompting can improve verification.

CodeBLEU

- New evaluation metric for code synthesis.
- Improves BLEU by considering code syntax and semantics.

$$\text{CodeBLEU} = \alpha \cdot \text{BLEU} + \beta \cdot \text{BLEU}_{\text{weight}} + \gamma \cdot \text{AST} + \delta \cdot \text{DataFlow}$$

BLEU

- Most important metric for machine translation evaluation

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

- BLEU score is the geometric mean of n-gram precisions.
- Typically $w_1 = \ldots = w_N = \frac{1}{N}$.

BLEU

- **Precision** is the ratio of n-grams in the candidate to the reference.

$$p_n = \frac{\sum_{C \in \text{Candidates}} \sum_{i=1}^{n} c_{\text{clip}}^{C}(i \ldots i+n)}{\sum_{C \in \text{Candidates}} \sum_{i=1}^{n} c^{C}(i \ldots i+n)}$$

- c_{clip}^{C} is the count of n-grams in the candidate that appear in the reference.

CodeBLEU - Weighted BLEU

- Uses weighted n-gram match for keywords.

$$p_n = \frac{\sum_{C \in \text{Candidates}} \sum_{i=1}^{n} \mu_n^i c_{\text{clip}}^{C}(i \ldots i+n)}{\sum_{C \in \text{Candidates}} \sum_{i=1}^{n} \mu_n^i c^{C}(i \ldots i+n)}$$

- $\mu_n^i = 5$ for keywords, 1 otherwise.

CodeBLEU - AST Match

- Leverages abstract syntax trees (ASTs) for structure.

$$\text{AST} = \frac{c_{\text{clip}}(T_{\text{cand}})}{c(T_{\text{ref}})}$$

CodeBLEU - Data-Flow Match

- Incorporates data-flow analysis for semantics.
- Difference between `return x` and `return y` is captured.
- Ensures logic correctness is evaluated.

CodeBLEU Evaluation

- Better correlation with human evaluation scores.
- Evaluated on text-to-code, translation, and refinement tasks.
- Surpasses BLEU and perfect accuracy in reliability.
- Hyperparameter tuning improves correlation.

Summary

- Code generation systems like StarCoder, REDCODER, and ChatGPT are evaluated.
- Techniques from large language modelling are applied to code generation.
- CodeBLEU is a new metric for code synthesis evaluation.